



Multimedia Security and Privacy

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Outline

Usages of Digital Watermarking

Watermarking vs Data Hiding

Watermark detection

Watermark detection in practice

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Watermarking

Definition

Ensures ownership and authenticity of digital content

Requirements

- Robustness
- Percebility
- Embedding Capacity ($\sim 64 - 128$ bits)

Applications

- Copyright protection
- Real vs synthetic media detection

Steganography

Definition

Hides the existence of a message for secret communication

Requirements

- Undetectability
- Embedding Capacity (~ 1 Ko per image)

Applications

- Alternative to cryptography

Tamper-proofing

Definition

Detects unauthorized modifications to digital content (Key based or compression-based)

Requirements

- Capacity
- Robustness
- Localization

Applications

- Integrity verification
- Forensics



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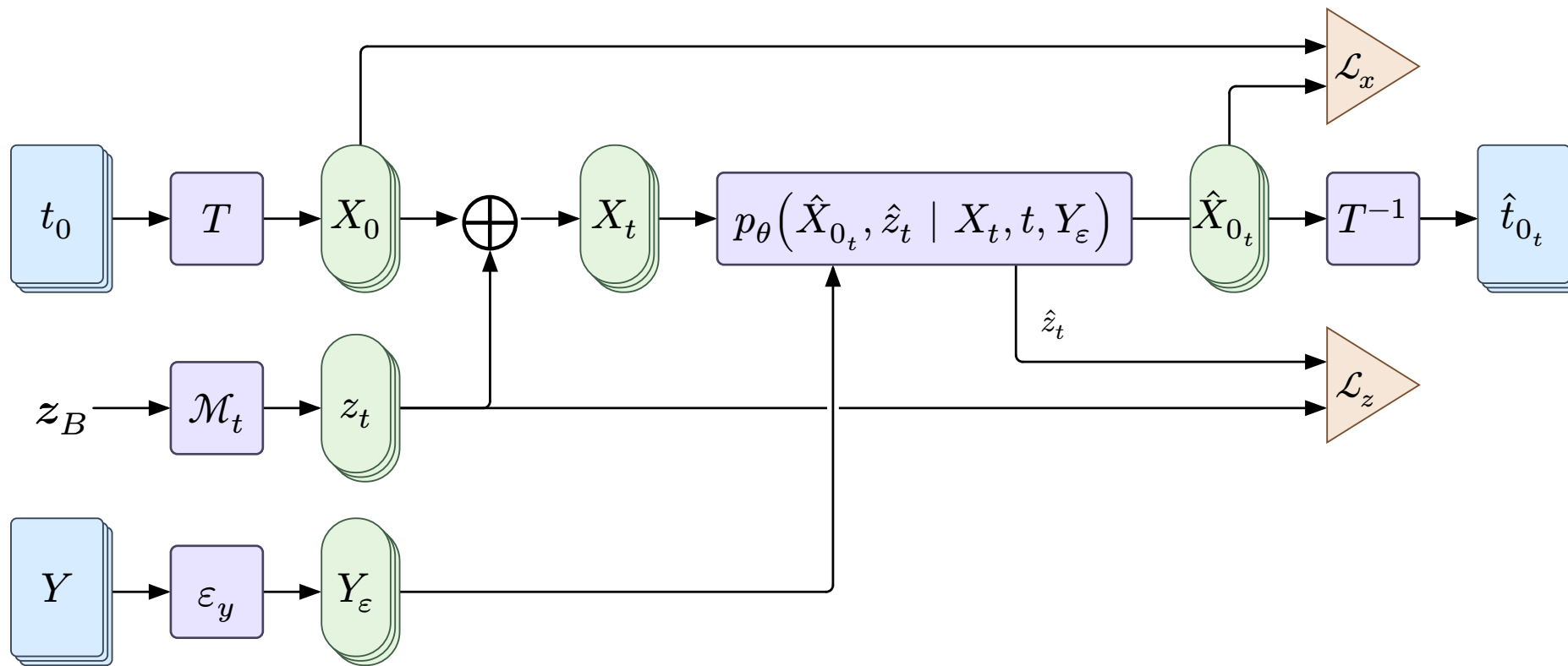
Watermarking vs Data Hiding

Technologies

- Zero-bit watermarking (Detection problem)
- Multi-bit watermarking (Decoding problem)

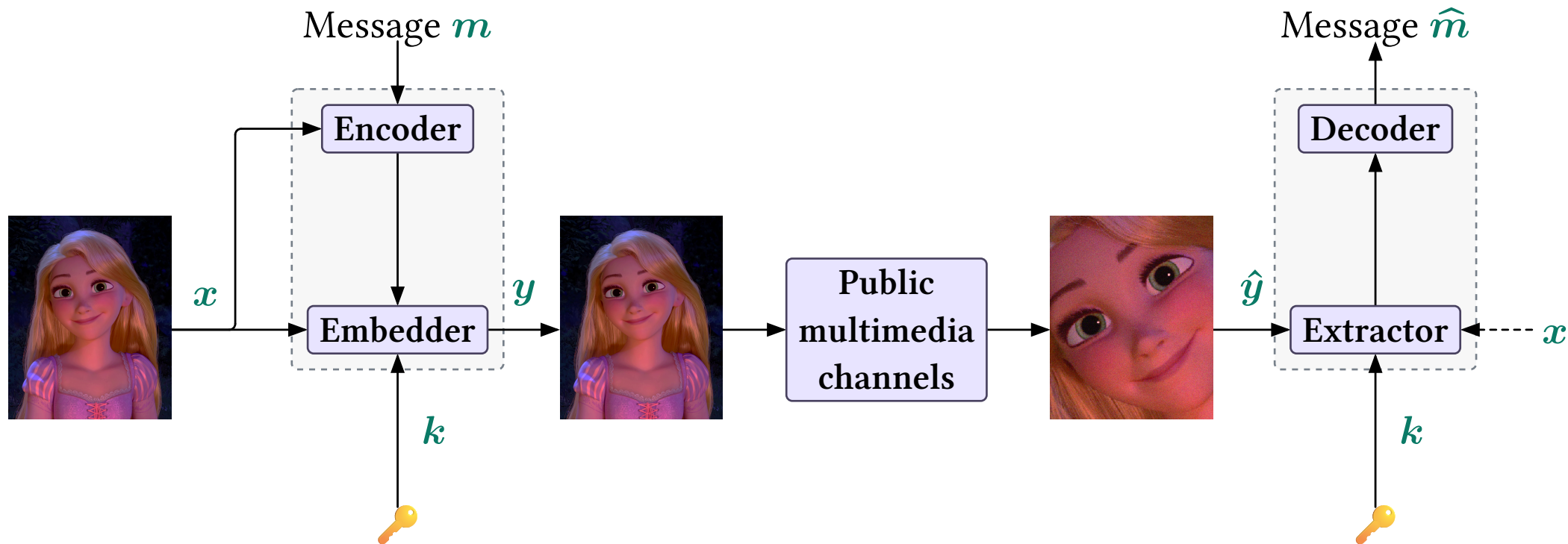


Block diagram





Block diagram



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Hypothesis testing

- $H_0 : y = x$
- $H_1 : y = x + w$

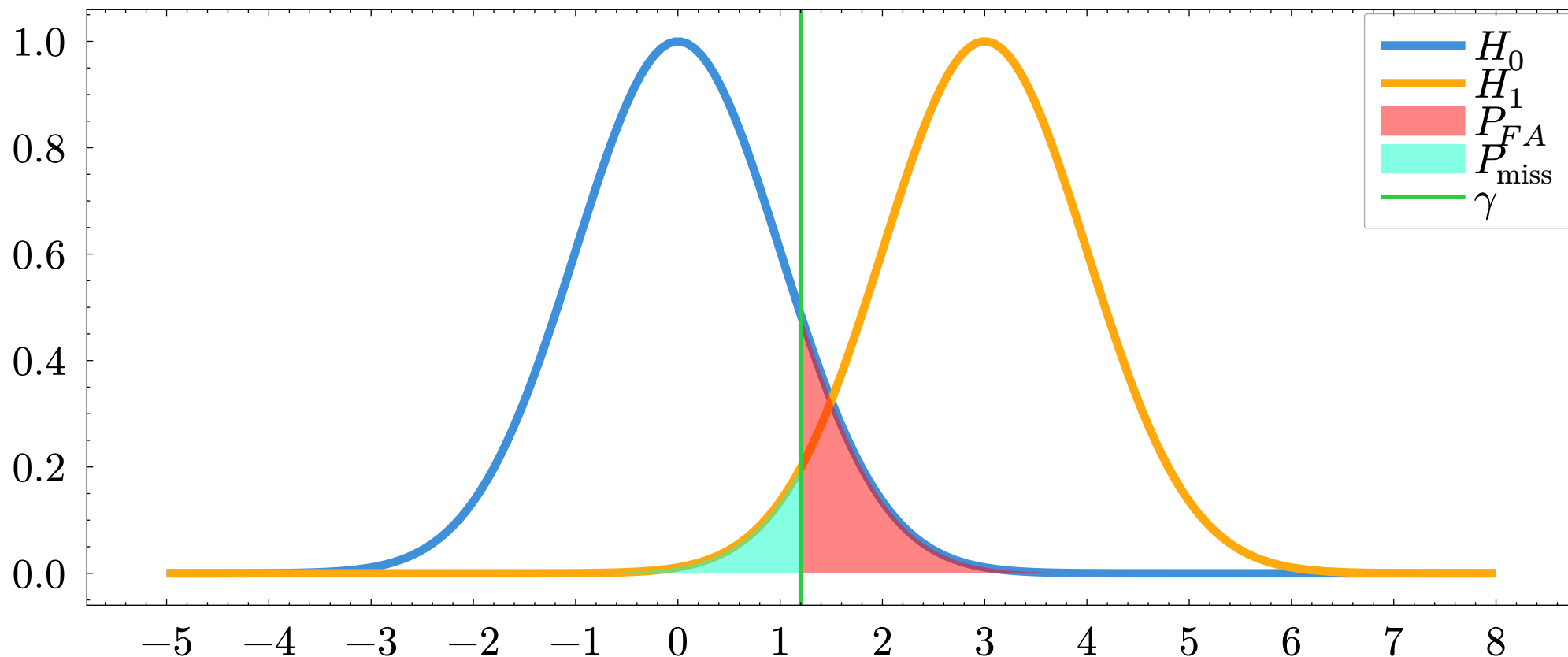
Score definition

- $S = \frac{y^T w}{M}$ (M : number of pixels)
- $S \sim \mathcal{N}(\mu_{H_0}, \sigma_{H_0}^2)$ under H_0 (Central Limit Theorem)
- $S \sim \mathcal{N}(\mu_{H_1}, \sigma_{H_1}^2)$ under H_1

Probabilities

- Correct Detection: $P_{D(\gamma)} = \Pr(S \geq \gamma \mid H_1)$
- False Acceptance: $P_{FA}(\gamma) = \Pr(S \geq \gamma \mid H_0)$

Graph hypothesis testing



Graph hypothesis testing

Neyman Pearson Strategy

- Fix P_{FA} :

$$P_{FA}(\gamma) = \int_{>\gamma} p(x \mid H_0) dx$$

- Minimize probability of miss:

$$P_{\text{miss}} = 1 - P_D = \int_{\leq \gamma} p(x \mid H_1) dx$$

Bayesian Strategy

- Fix priors $P(H_0)$ and $P(H_1)$
- Minimize overall error: $P_{\text{error}} = P_{FA}P(H_0) + P_{\text{miss}}P(H_1)$



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Cosine similarity

- $\cos(\theta) = \frac{y^T w}{\|y\|_2 \|w\|_2}$

Score definition

- $S(y) = \frac{1}{M} (y^T w) = \frac{1}{M} \|y\|_2 \|w\|_2 \cos(\theta)$

Distribution under H_0

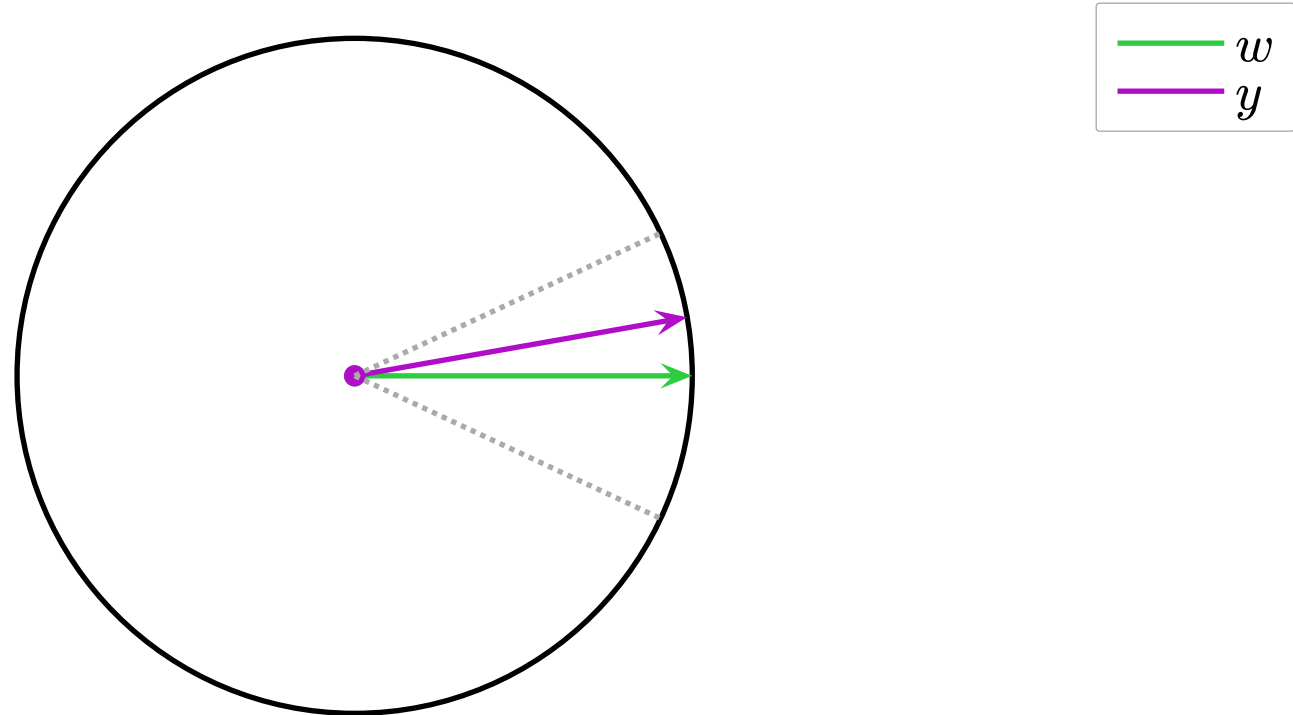
- $S(y) = \frac{1}{M} x^T w \sim \mathcal{N}\left(0, \frac{\|w\|_2^2 \sigma_x^2}{M^2}\right)$

Distribution under H_1

- $S(y) = \frac{1}{M} (x + w)^T w \sim \mathcal{N}\left(\frac{\|w\|_2^2}{M}, \frac{\|w\|_2^2 \sigma_x^2}{M^2}\right)$

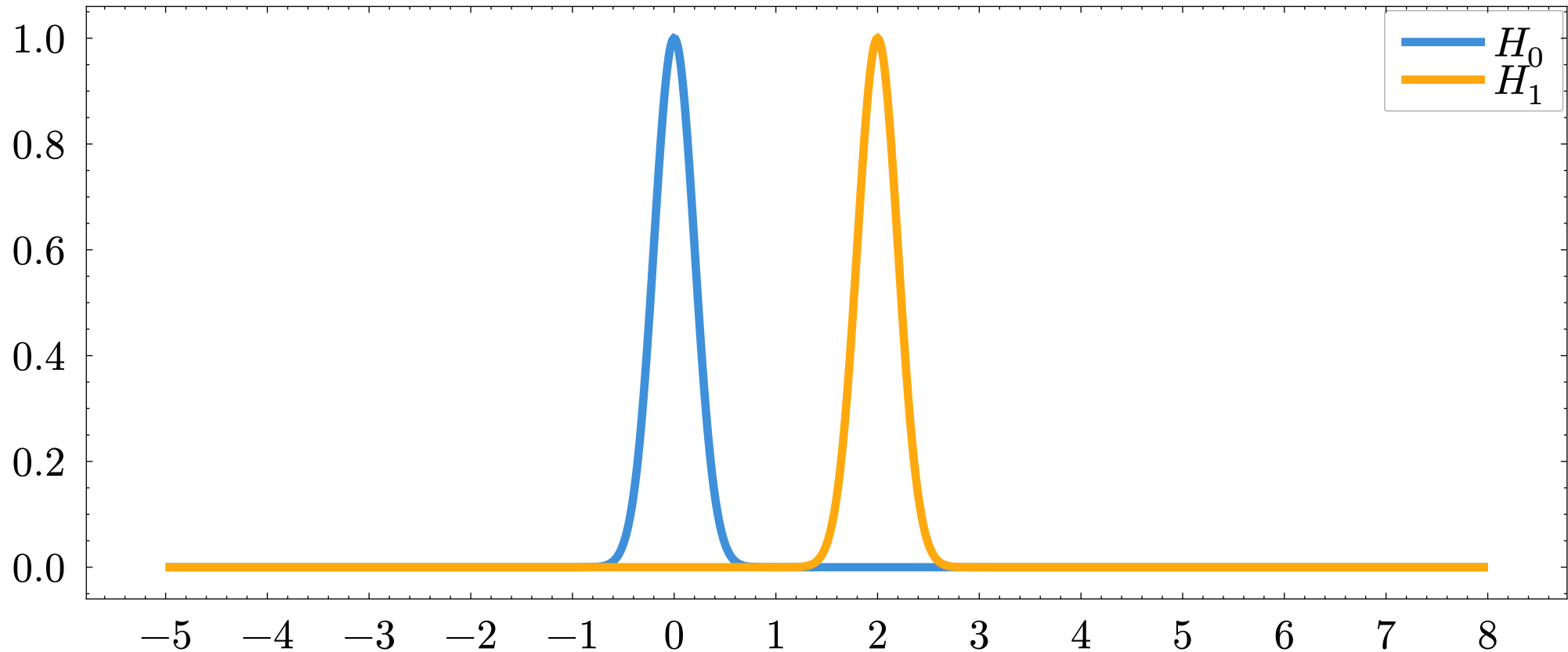


Decision rule: $|y^T w| > \|y\|_2 \|w\|_2 \cos(\theta)$



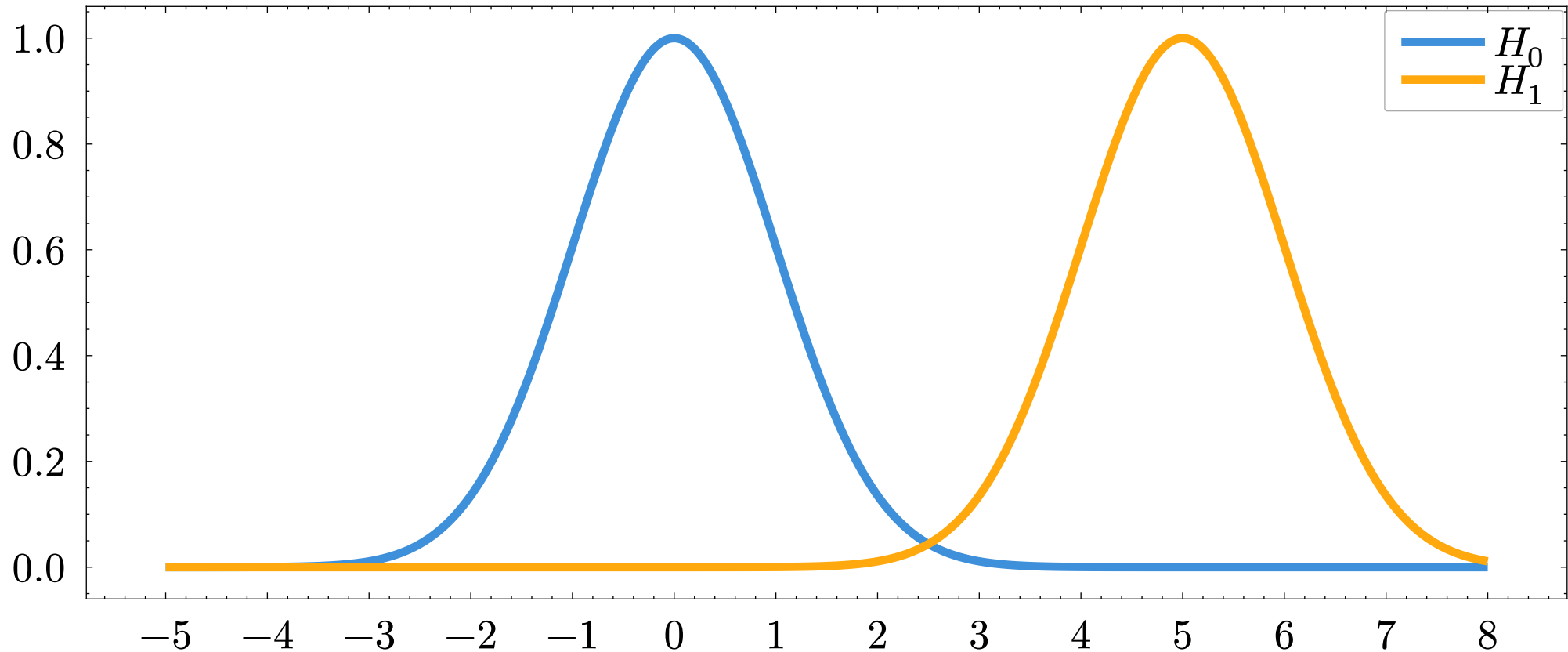


M increases





$\|w\|_2^2$ increases





σ_x^2 increases

